UNIVERSITY OF CAMBRIDGE

Retrieval Augmented Generation (RAG)

Focus of the day: RAG

- Introduction to RAG
- RAG Architecture
- RAG implementation exercise

INTRODUCTION TO RAG

RAG

- **RAG:** combines the benefits of search with LLMs ; ley advantages:
  - Enhances factual accuracy, thus reducing risks and challenges linked to LLMs “hallucinating”
  - Provides relevant contextual information
  - Enables safer usage of proprietary information and data
  - Control information permissions
  - Enables use of more uptodate info
  - Multi-modal

RAG Use Cases

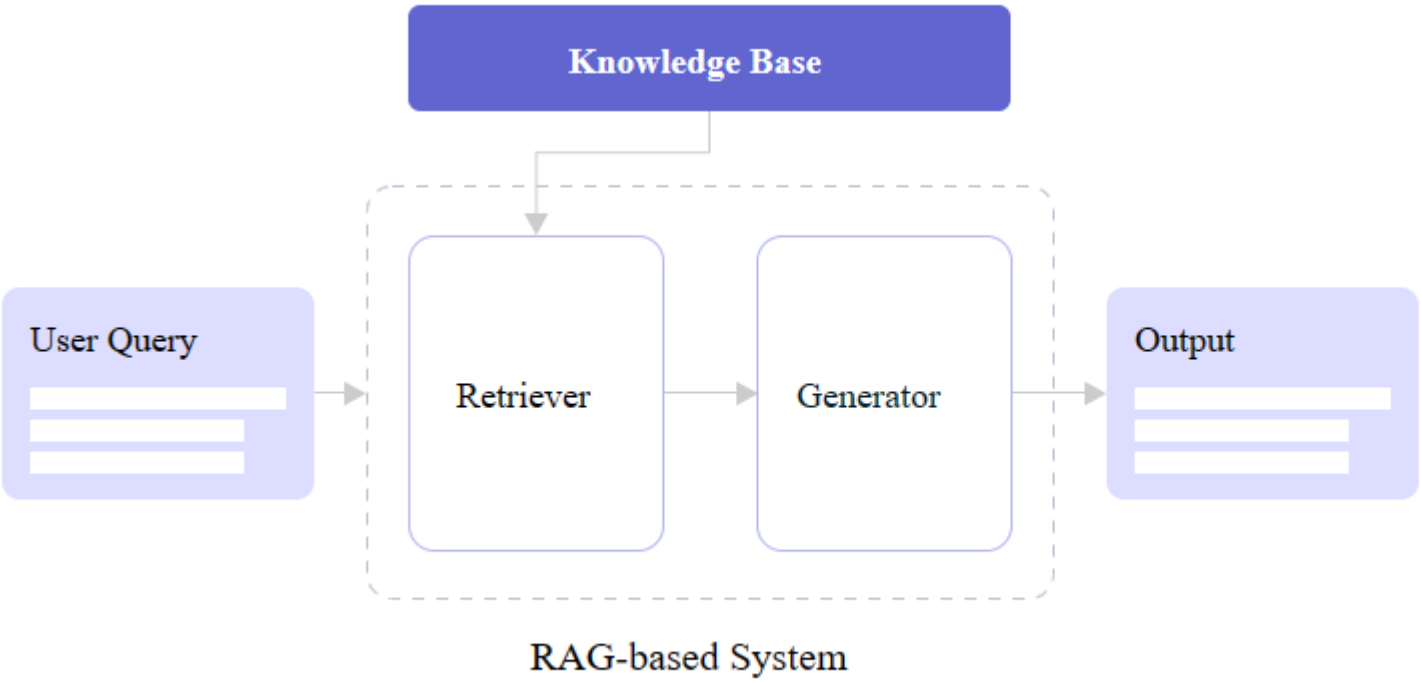
- Customer service for internal and external clients (e.g.HR)
- Q&A
- Research assistant

RAG ARCHITECTURE

RAG Architecture

- RAG application
  - Retrieve information using a vector database
  - Augmenting the original query (prompt) with relevant information
  - Generating an answer to the original query using an LLMs and the retrieved information

RAG architecture



Source: https://www.smashingmagazine.com/2024/01/guide-retrieval-augmented-generation-language-models/

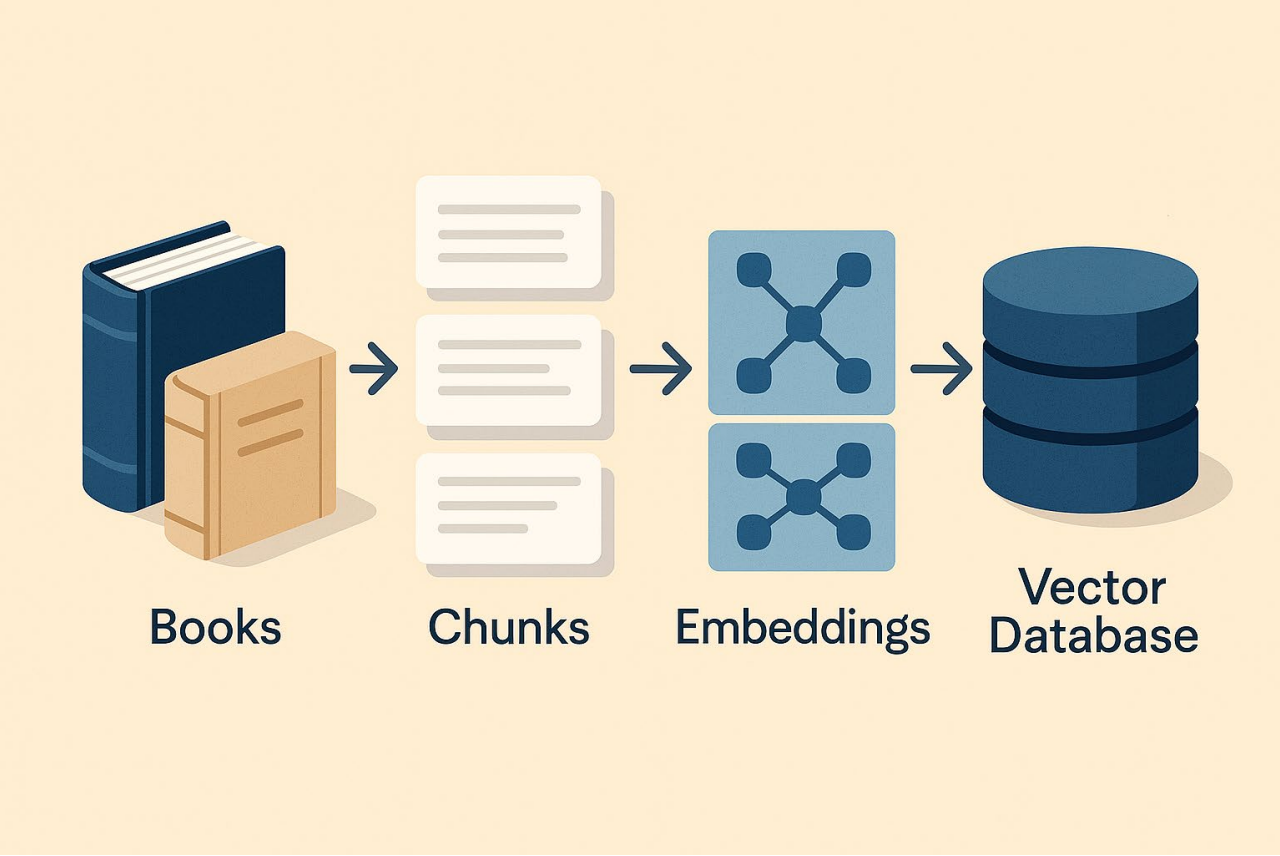
RAG: A closer look at the retrieval process

- Retrieval requires searching your knowledge base and extracting “relevant” text:
  - Keyword search (e.g. term frequency); But there are known issues(let’s refresh them)
  - Semantic search/Embeddings
  - Embeddings=numerical representations/”Coordinates on high-dimensional space
  - Embedding models take text and turn into numerical representation that represent meaning

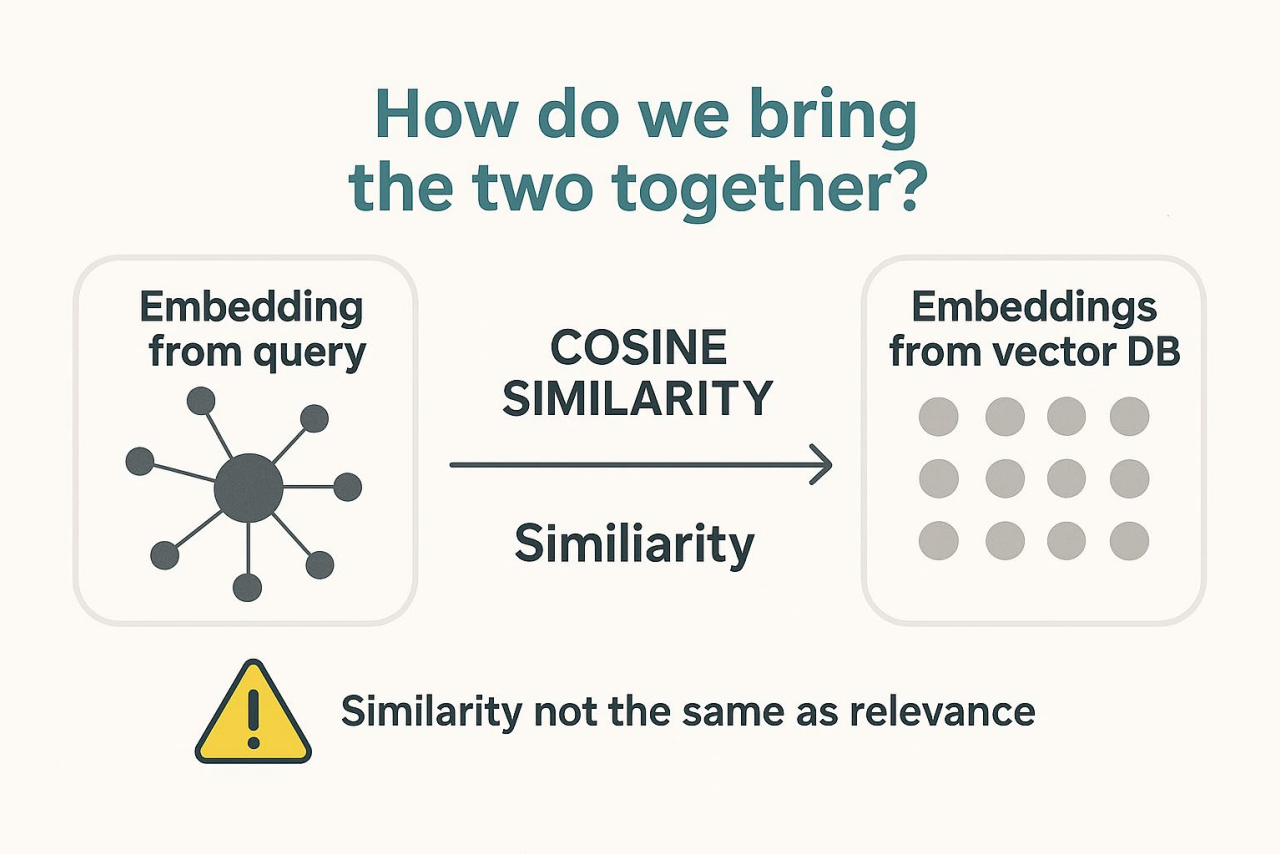
RAG: A closer look at the retrieval process

- Retrieval requires searching your knowledge base and extracting “relevant” text:
  - The retrieval process requires chunking “knowledge base” i.e. documents; chunking is critical because if your chunks are too small, you may miss context; if they are too big, you may lose precision
  - Chunking methods:
    - Fixed size chunks; simplest model; issues e.g. splitting a sentence.
    - Sentence chunking
    - Paragraph chunking
  - Good rule of thumb: 200-500 characters with a 50-100 overlap; avoid sentence or word splitting

RAG: A closer look at the retrieval process



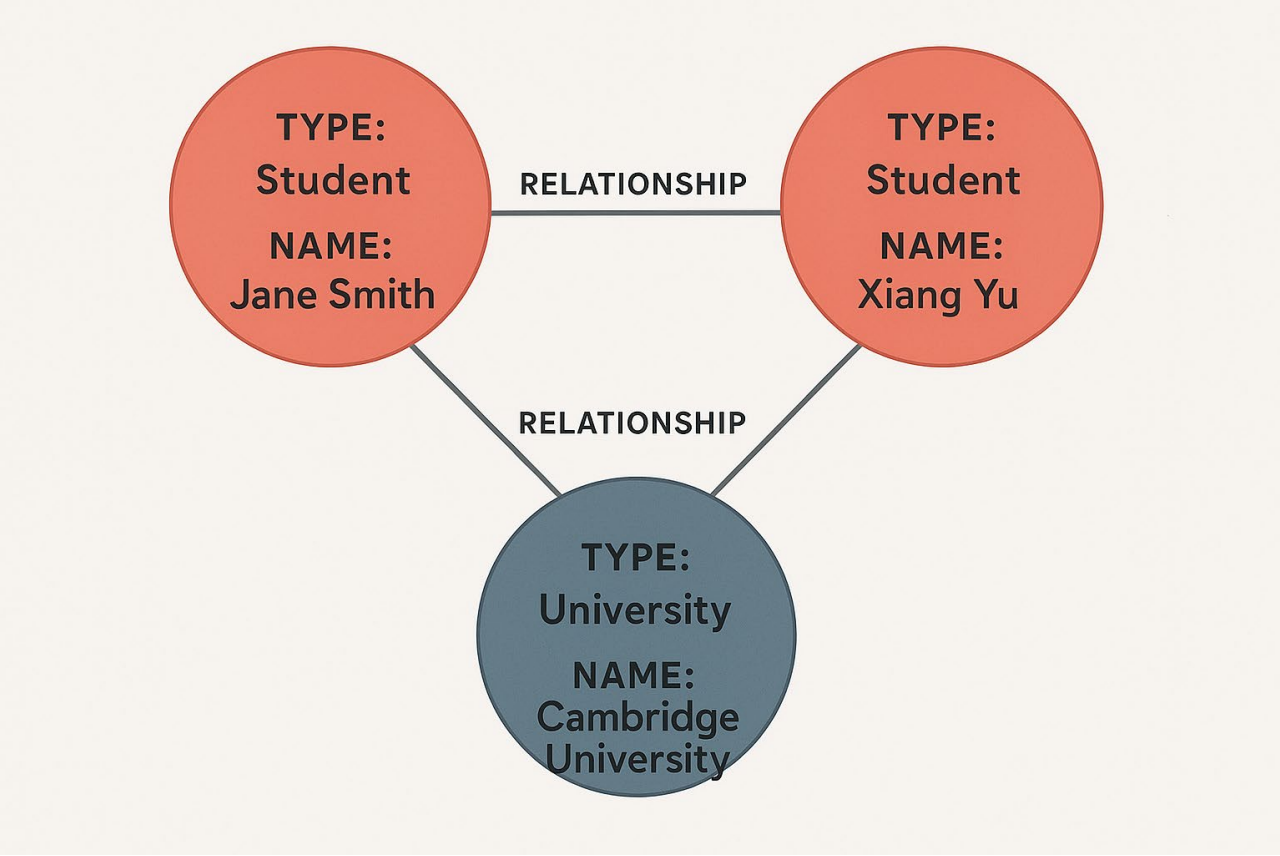
RAG: A closer look at the retrieval process



Graph RAG

- RAG as described previously-referred to as baseline RAG- works well for articles/emails/book but less effective when it is distributed across multiple documents i.e. cross-references between documents or implicit references; in other words, it works best with explicit mentions in the chunked text.
- Knowledge graphs represents info in a structured interconnected fashion- Key components: entities(nodes); relationships (edges) between entities.
  - ***Entities:*** real world objects, concepts, ideas;
  - ***Relationships***: ways they are connected
  - **Attributes:** additional properties of the entities
- ***Traversing relationships*** in a knowledge graph may make inferences possible when things are not explicitly stated.

Graph RAG


TYPE:
Student
NAME:
Jane Smith
RELATIONSHIP
TYPE:
Student
NAME:
Xiang Yu
RELATIONSHIP
TYPE:
University
NAME:
Cambridge
University

RAG approaches comparisons

Feature	Knowledge Graphs	Vector Databases
Data Representation	Entities (nodes) and relationships (edges) forming a graph structure.	High-dimensional vectors representing pieces of Information (e.g., document, sentence).
Retrieval Mechanisms	Traversing graph structure and following relationships. Enables inference and derivation of new knowledge.	Vector similarity based on similarity metric (e.g., cosine similarity). Returns most simirar vectors and associated information
Interpretability	Facilitates integration by representing entities and rationships in a unified graph structure. Seamiess integration if entities and relationships are mapped pro-	Less interpretable to hurnans due to high-dimensional numerical representations. Challenging to directly understand relationships or reasoning behind retrieved information.
Knowledge Integration	Enables inferential reasoning by traversing the graph structure and leveraging relationships between entities. Uncovers implicit connecns and derives new insights.	Limited, relying on vector similarity and may miss implicit relationships or inferences. Can identify similar information but not complex relationships from knowledge graphs.

CLASSROOM EXERCISES

Classroom exercises

- **Build a Knowledge Graph** https://github.com/DylanTartarini1996/knowledge-graphs
- **Baseline RAG example**: Please reproduce the example in https://huggingface.co/blog/ngxson/make-your-own-rag and evaluate it using different questions

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